Exercise 1

$f_1 = 3\ GHz$, $f_2 = 4{,}32\ GHz$

$IL > 15\ dB$ at $f_{R_1} = 2{,}2\ GHz$ and $f_{R_2} = 5{,}13\ GHz$

$Z_a = 50\ \Omega$

The idea is that we want to develop a low-pass equivalent filter, so the first thing to do to map band-pass specs to a low-pass prototype is to evaluate the center frequency (f_0) and the fractional bandwidth (Δ)

$$f_0 = \sqrt{f_1 \cdot f_2} = \sqrt{3 \cdot 4{,}32 \cdot 10^9 \cdot 10^9} = 3{,}6\ GHz$$

$$\Delta = \frac{f_2 - f_1}{f_0} = \frac{4{,}32 - 3}{3{,}6} = 0{,}3667 = 36{,}67\ \%$$

$$\omega_0 = 2\pi f_0 = 22{,}62 \cdot 10^9\ rad/s$$

Now we normalize the rejection frequencies:

- At f_{R_1}: $\tilde{\omega}_{R_1} = \frac{1}{\Delta}\left(\frac{f_{R_1}}{f_0} - \frac{f_0}{f_{R_1}}\right) = \frac{1}{0{,}3667}\left(\frac{2{,}2}{3{,}6} - \frac{3{,}6}{2{,}2}\right) = -2{,}796$

- At f_{R_2}: $\tilde{\omega}_{R_2} = \frac{1}{\Delta}\left(\frac{f_{R_2}}{f_0} - \frac{f_0}{f_{R_2}}\right) = \frac{1}{0{,}3667}\left(\frac{5{,}13}{3{,}6} - \frac{3{,}6}{5{,}13}\right) = 2{,}028$

Since the IL must be greater than 15 dB at both frequencies and since in the normalized domain the response is symmetrical, we can use the smallest $\tilde{\omega}$ (in absolute value) to evaluate the filter order.

For a Butterworth filter $\Rightarrow IL = 10 \log_{10}(1 + \tilde{\omega}^{2N})$

$$15 \leq 10 \log_{10}(1 + 2{,}028^{2N})$$

$$10^{1{,}5} \leq 1 + 2{,}028^{2N} \Rightarrow 31{,}62 \leq 1 + 2{,}028^{2N} \Rightarrow 30{,}62 \leq 2{,}028^{2N}$$

$$2N \log(2{,}028) \geq \log(30{,}62) \Rightarrow 2N \geq \frac{\log(30{,}62)}{\log(2{,}028)} \Rightarrow N \geq 2{,}42$$

Since N has to be an integer, $N = 3$.

Looking up the Butterworth table we can see that $g_1 = g_3 = 1{,}0$, $g_2 = 2{,}0$ and the layout of the network is: series branch, shunt branch, series branch.

- Section 1

$$L_1 = \frac{g_1 Z_a}{\Delta \omega_0} = \frac{1{,}0 \cdot 50}{0{,}3667 \cdot 22{,}62 \cdot 10^9} = 6{,}029\ nH$$

$$C_1 = \frac{\Delta}{g_1 Z_0 \omega_0} = \frac{0,3667}{1,0 \cdot 50 \cdot 22,62 \cdot 10^3} = 0,324 \text{ pF}$$

- Section 2

$$C_2 = \frac{g_2}{\Delta Z_0 \omega_0} = \frac{2,0}{0,3667 \cdot 50 \cdot 22,62 \cdot 10^3} = 4.823 \text{ pF}$$

$$L_2 = \frac{\Delta Z_0}{g_2 \omega_0} = \frac{0,3667 \cdot 50}{2,0 \cdot 22,62 \cdot 10^3} = 0,405 \text{ uH}$$

- Section 3 = Section 1

$L_3 = L_1 = 6,029$ uH

$C_3 = C_1 = 0,324$ pF

Exercise 2

We design the filter of exercise 1 by using $\lambda/4$ shunt stubs.

$\varepsilon_r = 2,54$, $h = (1/8)'' = 3,175$ mm

Width between 0,5 and 1,5 ?

By using the parameters $\Delta = 0,3667$ and $Z_0 = 50$, we can evaluate the impedence of the shunt sectors

$$Z_{0_1} = \frac{\pi}{4} \frac{Z_0 \Delta}{g_1} = \frac{\pi}{4} \frac{50 \cdot 0,3667}{1,0} = 14,4 \ \Omega$$

$$Z_{0_2} = \frac{\pi}{4} \frac{Z_0 \Delta}{g_2} = \frac{\pi}{4} \frac{50 \cdot 0,3667}{2,0} = 7,2 \ \Omega$$

$$Z_{0_3} = \frac{\pi}{4} \frac{Z_0 \Delta}{g_3} = \frac{\pi}{4} \frac{50 \cdot 0,3667}{1,0} = 14,4 \ \Omega$$

As we can see these are very low impedance traces, which usually means that they have to be pretty large, let's see if they can be manifactured with the spec.

We evaluate for the worst case, in this case the second shunt, that has the lowest resistance.

$$Z_{0_2} = \frac{120\pi}{\sqrt{\varepsilon_r} \ W/h} \Rightarrow \frac{W}{h} = \frac{120\pi}{\sqrt{\varepsilon_r} \ Z_{0_2}} \simeq \frac{237}{7,2} \simeq 32,9$$

Given that $h = 3,175$ mm $\Rightarrow \frac{W}{h} = 32,9 \Rightarrow W = 32,9 \cdot 3,178 = 104$ mm

So we can't meet the specification.

Exercise 3

To design a BPF using coupled lines we need an N+1 array of coupled lines sections. Each one acts as an inverter J and is quarter wave long at f_0.

electrical length = 90°

Knowing that:

$g_1 = 1$

$g_2 = 2$

$g_3 = 1$

$g_4 = 1$

We can calculate even and odd impedances for each section but first we need to evaluate the Admittance Inverter Constants.

- Section 1 and 4 (outer sections)

$$Z_0 J_1 = Z_0 J_4 = \sqrt{\frac{\pi \delta}{2 g_1 g_4}} = \sqrt{\frac{\pi \cdot 0{,}3667}{2 \cdot 1 \cdot 1}} = 0{,}7589$$

- Section 2 and 3 (inner sections)

$$Z_0 J_2 = Z_0 J_3 = \sqrt{\frac{\pi \delta}{2 g_2 g_3}} = \sqrt{\frac{\pi \cdot 0{,}3667}{2 \cdot 2 \cdot 1}} = 0{,}4073$$

Knowing that:

$$Z_{ie} = Z_0 [1 + J Z_0 + (J Z_0)^2]$$

$$Z_{io} = Z_0 [1 - J Z_0 + (J Z_0)^2]$$

- For the outer sections

$$Z_{1e} = Z_{4e} = 50 [1 + 0{,}7589 \cdot 50 + (0{,}7589 \cdot 50)^2] = 116{,}75\,\Omega$$

$$Z_{1o} = Z_{4o} = 50 [1 - 0{,}7589 \cdot 50 + (0{,}7589 \cdot 50)^2] = 40{,}85\,\Omega$$

- For the inner sections

$$Z_{2e} = Z_{3e} = 50 [1 + 0{,}4073 \cdot 50 + (0{,}4073 \cdot 50)^2] = 78{,}66\,\Omega$$

$$Z_{2o} = Z_{3o} = 50 [1 - 0{,}4073 \cdot 50 + (0{,}4073 \cdot 50)^2] = 37{,}93\,\Omega$$

By using the website we get that:

- for sections 1 and 4

$W = 7{,}0383$ mm

$S = 0{,}2563$ mm

$L = 577{,}213$ mm

- for sections 2 and 3

$W = 6{,}527$ mm

$S = 0{,}391$ mm

$L = 578{,}055$ mm